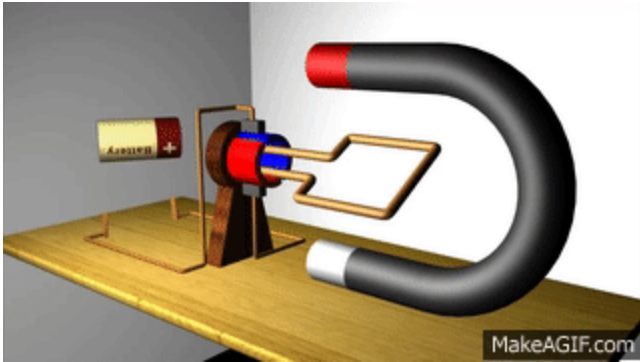


16-popquiz loop in magentic field, magnetic force on particle, current carrying loop $\updownarrow$

⚠ This is a preview of the published version of the quiz

Started: Mar 22 at 7:14pm

Quiz Instructions



Question 1 17 pts

A neutron (no charge) is moving @ right in a magnetic field (going toward you, out of the screen). The force on the neutron



is @ down



is 0



I missed the class

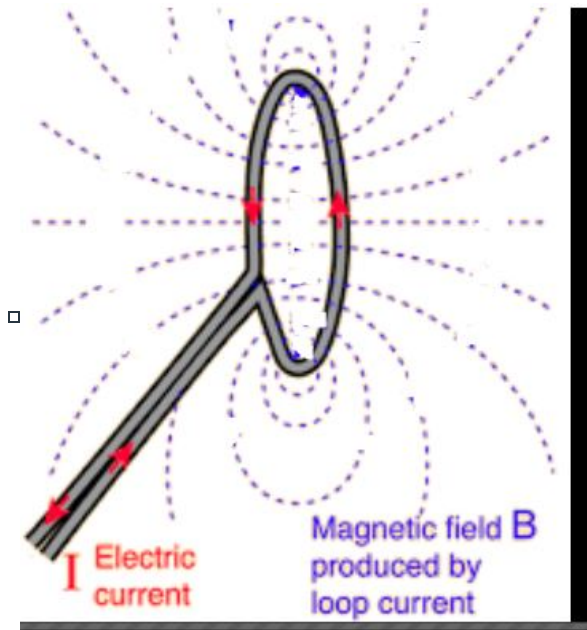


is @ up



Question 2 17 pts

A loop of current carrying loop is shown below. The loop is perpendicular to your screen and the current is going CCW. Use the right hand rule (last class). The magnetic field


☐

from left to right. the Left side of the loop is the North

☐

I missed the class

☐

flows from right to left. The right side of the loop is the South

☐

The right side of the loop is the North, It flows from N to S

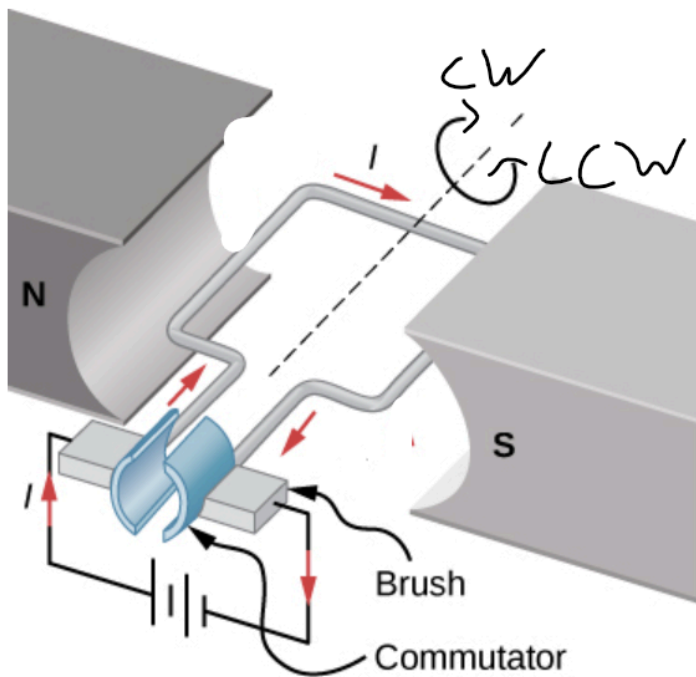
⋮

Question 3 17 pts

a current carrying loop inside a magnetic field as shown. What is true (use your right hand)

hint : you have 2 ways to do it. with your palm curl around the loop , fingers follow the current. the thumb is the magnetic field of the loop and wants to align itself with the magnetic field of the magnet (N to S)

or use right hand to draw the magnetic forces applied to the side of the loop (length). and you will know the direction of the rotation (the forces provide a torque)



- ☐ the loop rotates CW
- ☐ the loop rotates CCW
- ☐ the loop does not rotate
- ☐ I don't have a right hand



Question 4 17 pts

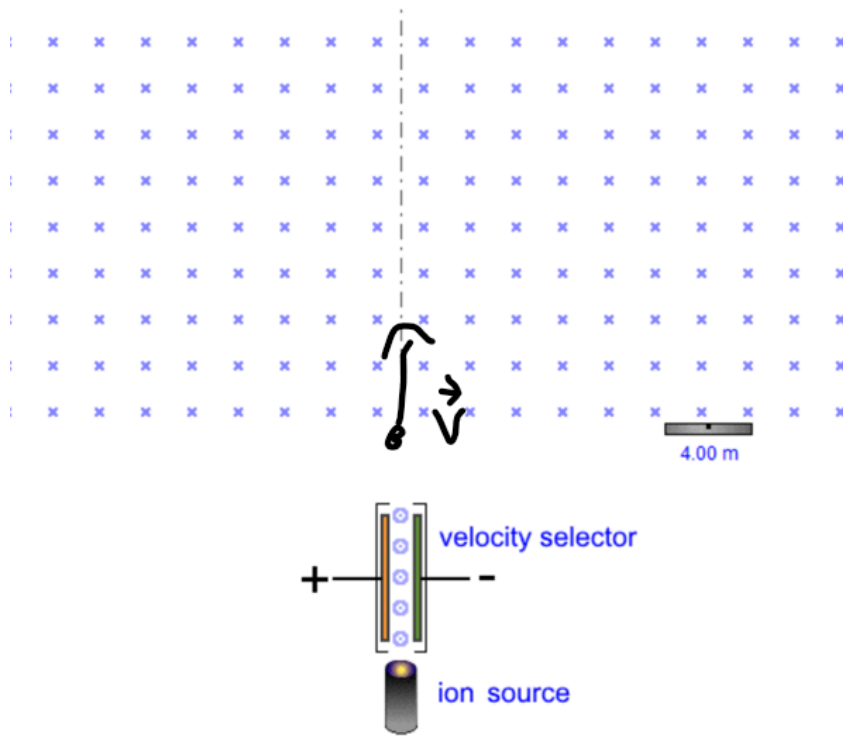
A proton is moving to the right (in the plane of this screen) inside a magnetic field which is @ left (in the plane of this screen). So the velocity of the proton and the field have anti parallel directions.

The force acting on the proton is

- ☐ up (in the plane of this screen)
- ☐ down (in the plane of this screen)
- ☐ The force is away from you. Into the screen X
- ☐ 0 (no force)



Question 5 17 pts



A proton is accelerated in an electric field and enters a magnetic field with a velocity V . The magnetic field is uniform and going into the screen (crosses). The proton will d

☐

Will speed up along a straight line

☐

describe half a circle and will be deflected @ left (of the dash line)

☐

describe half a circle and will be deflected @ right (of the dash line)

☐

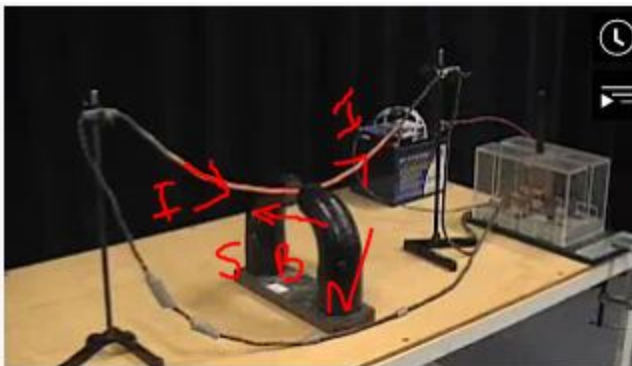
Will keep moving in a straight line at a constant speed

⋮

Question 6 17 pts

A current carrying-wire is placed in the magnetic field of a horse shoe magnet as shown below.

The force acting on the wire





is up



there is no force



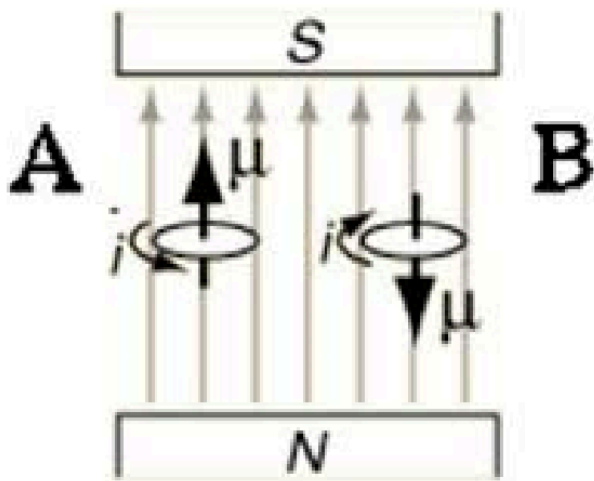
I missed the class



is down



Question 7 4 pts



In this setup, you see two loops of current in an uniform magnetic field. The loop wants



to be in position A (largest state of energy)



keeps spinning about its axis for fun



to be in position A (lowest state of energy)

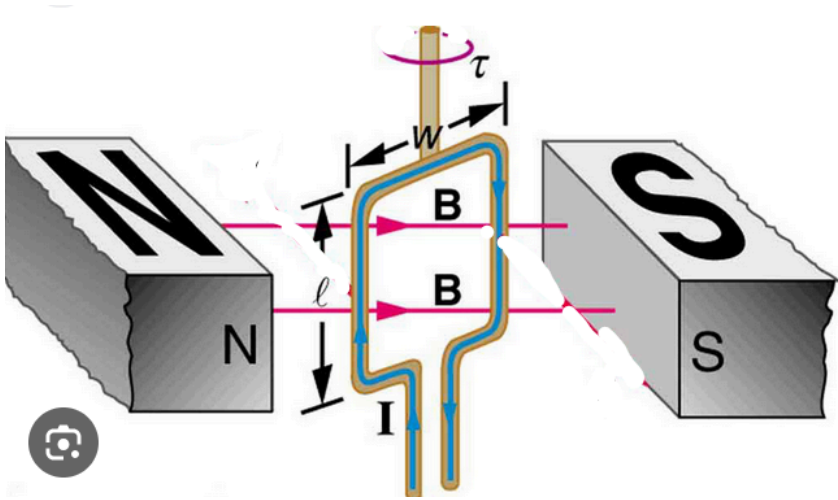


to be in position B (lowest state of energy)



Question 8 4 pts

bonus point:



The magnetic forces applied to the loop provide a torque. The loop

- ☐ spins CCW
- ☐ I really hate Physics and don't care
- ☐ does not move
- ☐ spins CW



Question 9 3 pts

MCAT bonus

A magnetic field circulates around charged particles. use the right hand. the thumb is the velocity of the positive charge and the fingers curl around. But for a negative charge. the magnetic field curls in the opposite direction.

So for a negative charge, use your left hand. your thumb is the velocity of the negative charge. the magnetic field is the second finger. the third finger shows you which way its curving

A moving negative charge placed in an external magnetic field circulates counterclockwise in the plane of the paper. In which direction is the magnetic field pointing?

- A. Into the page
- B. Out of the page
- C. Toward the center of the circle
- D. Tangent to the circle



C



D



B



A

Not saved

Submit Quiz